Mechanical Waves and Sound Waves

Mechanical Waves

Definition

Mechanical waves are waves that propagate through a material medium due to the vibration of its particles.

They cannot travel in a vacuum.

Examples

- Water waves
- Waves on a rope
- Sound waves (a special type of mechanical wave)

Types of Mechanical Waves

1. Transverse Waves

- Particle motion is perpendicular to the direction of wave propagation
- Example: waves on a string

2. Longitudinal Waves

- Particle motion is parallel to the direction of propagation
- Example: sound waves in air

**Wave Behaviors

*Reflection

Wave bouncing back from a surface

*Refraction

Change in direction when a wave enters a different medium

*Diffraction

Spreading of waves around obstacles

***Interference**

Superposition of two or more waves

Sound Waves

Definition

Sound waves are longitudinal mechanical waves produced by vibrating objects and transmitted through a medium.

Nature of Sound Waves

- Longitudinal waves
- Consist of:
- Compressions
- Rarefactions

Properties of Sound

1. Intensity

Related to loudness

2. Pitch (Frequency)

- High frequency → high pitch
- Low frequency → low pitch

3. Timbre

Quality that distinguishes different sounds with the same pitch

Types of Sound Waves (by frequency)

Type	Frequency
Infrasonic	< 20 Hz
Audible	20 – 20,000 Hz
Ultrasonic	> 20,000 Hz

Applications of Sound Waves

- SONAR systems
- Medical imaging (ultrasound)
- Depth measurement
- Communication systems

Sound Waves in Liquids and Solids

1. Sound Waves in Liquids

1.1 Nature of Propagation

- In liquids, sound waves are longitudinal waves only.
- Particle motion is parallel to the direction of wave propagation.
- The wave consists of compressions and rarefactions.

1.3 Characteristics

- Speed of sound in liquids is higher than in gases.
- Depends on:
    - Density of the liquid
    - Compressibility (bulk modulus)

- Example: Sound travels faster in water than in air.

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2. Sound Waves in Solids

2.1 Types of Waves in Solids

In solids, sound waves can propagate in two forms:

(a) Longitudinal Waves

- Particle vibration is parallel to wave direction.
- Similar to waves in gases and liquids.

(b) Transverse (Shear) Waves

- Particle vibration is perpendicular to wave direction.
- Possible only in solids due to strong intermolecular forces.

2.2 Reason for Wave Types in Solids

- Solids have rigid structure and strong restoring forces.
- They can resist both compression and shear stress.

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3. Speed of Sound in Solids

- Sound travels fastest in solids.
- Reasons:
    - Particles are tightly packed.
    - Strong bonding between particles.
    - Efficient energy transfer between particles.

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5. Comparison of Sound Propagation

Medium	Type of Waves Allowed	Speed of Sound
Gases	Longitudinal only	Slowest
Liquids	Longitudinal only	Medium
Solids	Longitudinal + Transverse	Fastest

Acoustic impedance

Definition

It is defined as the ratio of acoustic pressure to particle velocity in a medium:

$$Z=\frac{P}{V}$$

Where:

- (Z) = acoustic impedance
- (p) = sound pressure (Pa)
- (v) = particle velocity (m/s)

 For plane waves in a uniform medium:

$$Z=P*C$$

Where:

- (P) = density of the medium (kg/m^3)
- (c) = speed of sound in the medium (m/s)

 Why it matters (very important)

When a sound wave hits a boundary between two materials with different impedances:

- Part of the wave is reflected
- Part is transmitted

The greater the mismatch in impedance, the stronger the reflection.

***** Applications**

- Medical ultrasound imaging (detecting tissues, organs)
- Acoustics engineering (speaker design, soundproofing)
- Seismology (Earth layer analysis)
- Non-destructive testing (finding cracks in materials)